MTH 1125-Final Exam- NA

Exam time: 120 minutes
Date: June 8, 2026

NAME:
STUDENT ID NUMBER (Troy ID):
General instructions:

- No discussion is allowed in the exam room.
- No electronic devices (except calculator) are allowed.

PROBLEM	POINTS	SCORE
A1	4	
A2	4	
A3	4	
A4	4	
A5	4	
A6	4	
A7	4	
A8	4	
A9	4	
A10	4	
B1	10	
B2	15	
B3	15	
B4	10	
B5	10	
TOTAL	100	

PART A: Multiple choice or Short answer

(4 mark for each correct answer/choice)

A1. Find an equation of the slant asymptote of the curve $y = \frac{4x^3-3x^2+x-3}{2x^2-x}$.
Write yours answer here:

A2. Evaluate $\lim_{x\to 0} \frac{\sinh x - x}{x^3}$
Write the exact value here:

A3. Find the average value of the function $f(x) = |x^2 - 1| + x$ on the interval $[-2, 1]$.
Write your exact answer here:

A4. Let a, b be the length and width of a rectangle with perimeter 60 m whose area is as large as possible. Find $a - b$?
Circle the best answer.
A. 3
B. 1
C. 0
D. A,B,C are incorrect.

A5. Find the vertical asymptotes of $y = \frac{x^2-2x}{(x^2-4)(x+5)}$.
Write yours answer here:

A6. Evaluate the indefinite integral $\int \sqrt{x} \sin\left(1 + x^{\frac{3}{2}}\right) dx$.
Write yours answer here:

A7. When a particle is located a distance x meters from the origin, a force of $2x + 3$ newtons acts on it. How much work is done in moving the particle from $x = 0$ to $x = 2$?
Write yours answer here:

A8. Evaluate
$\lim_{x\to\infty}(e^x + 1)^{\frac{2}{x}}$
Circle the best answer.
A. 0
B. ∞
C. e^2
D. A,B,C are incorrect

A9. A bacteria culture grows with constant relative growth rate. The bacteria count was 400 after 2 hours and 3600 after 4 hours Find the number of cells after 5 hours
Circle the best answer.
A. 10800

B. 9900
C. 12000
D. A,B,C are incorrect

A10. Let R be the bounded region by $y = 16x - x^5, y = 0, 0 \leq x \leq 2$. Find the volume of the solid obtained by rotating R about y-axis.
Write the exact value here:

PART B: WRITE YOUR FULL ANSWERS.

B1. (a) Evaluate the Riemann sum for $f(x) = 4x^2 - 2x + 3$, taking the sample points to be right endpoints and $a = 0, b = 2$, and $n = 4$.
(b) Evaluate the Riemann sum for $f(x) = 4x^2 - 2x + 3$, taking the sample points to be the midpoints and $a = 0, b = 2$, and $n = 4$.

B2. (a) Find the derivative of the function
$h(t) = \int_t^{2t} e^{-x^2} dx.$
(b) Evaluate $\lim_{x \to 0} x^{-1} \int_x^{2x} e^{-t^2} dt.$

B3. (a) Sketch the bounded region R enclosed by the given curves and find its area: $y = x^2$ and $y = 7x$.
(b) Find the volume of the solid obtained by rotating R about $y = 3$
(c) Find the volume of the solid obtained by rotating R about y-axis.

B4. Evaluate
$\lim_{n \to \infty} \left(\frac{1}{2\sqrt{n^2+n}} + \frac{1}{2\sqrt{n^2+2n}} + \cdots + \frac{1}{2n\sqrt{2}} \right)$

B5. Find a continuous function f and a number a such that
$\int_a^x (7 + 2\sin t) f(t) dt = 2x + 10.$

------Good luck----
Start to write you solution (Part B).